Build a News ETL Data Pipeline Using Python and SQLite

In this project, we'll build a complete **ETL pipeline** (extract, transform, load) that retrieves real-time news data from the **News API**, transforms it from **semi-structured JSON** into a **structured format**, and loads it into an **SQLite database** for analysis. ETL processes are fundamental to data engineering and data integration, ensuring data is clean, consistent, and ready for business intelligence and analytics. We'll automate the entire data pipeline using **Apache Airflow** for scheduled execution and workflow orchestration.

We'll start by connecting to the **News API** to extract news articles in JSON format, then implement data transformation techniques to clean author columns, normalize fields, and convert the semi-structured data into structured tabular format using Pandas. Next, we'll design an SQLite database schema, create tables, and load the transformed data using SQL insert operations. We'll verify data integrity by querying the SQLite database and confirming successful data loading.

Finally, we'll automate the ETL workflow with Apache Airflow by initializing a **DAG (Directed Acyclic Graph)**, creating task operators for extraction, transformation, and loading stages, and implementing XComs for data passing between tasks. We'll configure the Airflow webserver, schedule the pipeline for regular execution, and implement error handling and best practices for production data pipelines.

By the end, we'll have a production-ready automated ETL system demonstrating Python data engineering, API data extraction, Pandas data transformation, SQLite database operations, Apache Airflow orchestration, and pipeline automation applicable to any data warehousing or data integration project.

The final implementation of the project will transform data from an unstructured format to a structured one, as illustrated below.

```
{
    "source": {
    "id": "fox-news",
    "name": "Fox News"
    },
    "author": "Michael Dorgan",
    "title": "Three staff stabbed at New Jersey hospital following fight between visitors",
    "description": "Three hospital staff members were stabbed at Newark Beth Israel Hospital in New Jersey after a fight broke out inside the pediatric ICU between two visitors.",
    "url": "https://www.foxnews.com/us/3-staff-stabbed-new-jersey-hospital-following-fight-visitors",
    "urlToImage": "https://static.foxnews.com/foxnews.com/content/uploads/2023/12/New-Jersey-Hospital-2.jpg",
    "publishedAt": "2023-12-08T19:38:17Z",
    "content": "Two nurses and a medical student were stabbed inside a New Jersey hospital's pediatric intensive care unit during a dispute between two visitors Friday morning.\r\nThe incident took place at around 7 a… [+1251 chars]"
    }
```


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Final Thoughts

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